ORF 524 — Final Exam

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Contents

Notes:

- Duration of examination: 180 minutes.
  An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
  Please provide as much detail as possible in your answers.
  Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
  A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
  This is otherwise a closed-book and individual examination.

1 Mean Square Error Prediction (25 points)

Assume Y, W, X be random variables with $\mathbb{E}[Y^2] < \infty$, $\mathbb{E}[W^2] < \infty$. Let $\mathcal{G} = \{g : \mathbb{E}[|g(X)|^2] < \infty\}$.

1. Show that

$$\mathbb{E}[(Y - g_1(X))(W - g_2(X))] = \mathbb{E}[(Y - \mathbb{E}[Y|X])(W - \mathbb{E}[W|X])] \\ + \mathbb{E}[(\mathbb{E}[Y|X] - g_1(X))(\mathbb{E}[W|X] - g_2(X))],$$

for all $g_1, g_2 \in \mathcal{G}$.

Solution: We have

$$\mathbb{E}[(Y - \mathbb{E}[Y|X] + \mathbb{E}[Y|X] - g_1(X))(W - \mathbb{E}[W|X] + \mathbb{E}[W|X] - g_2(X))] \\ = \mathbb{E}[(Y - \mathbb{E}[Y|X])(W - \mathbb{E}[W|X])] + \mathbb{E}[(\mathbb{E}[Y|X] - g_1(X))(\mathbb{E}[W|X] - g_2(W))],$$

because, for all $g \in \mathcal{G}$,

$$\mathbb{E}[(Y - \mathbb{E}[Y|X])(\mathbb{E}[W|X] - g(X))] = \mathbb{E}[\mathbb{E}[(Y - \mathbb{E}[Y|X])|X](\mathbb{E}[W|X] - g(X))] = 0 \\ \mathbb{E}[(\mathbb{E}[Y|X] - g(X))(W - \mathbb{E}[W|X])] = \mathbb{E}[(\mathbb{E}[Y|X] - g(X))\mathbb{E}[(W - \mathbb{E}[W|X])|X]] = 0.$$

2. Show that

$$\mathbb{E}[\mathbb{V}[Y|X]] = \min_{g \in \mathcal{G}} \mathbb{E}[(Y - g(X))^2].$$

Solution: From the first part, setting $Y = W$, it follows that $\mathbb{E}[Y|X] = \arg\min_{g \in \mathcal{G}} \mathbb{E}[(Y - g(X))^2]$, and therefore $\mathbb{E}[(Y - \mathbb{E}[Y|X])^2] = \mathbb{E}[\mathbb{E}[(Y - \mathbb{E}[Y|X])^2|X]] = \mathbb{E}[\mathbb{V}[Y|X]]$.

3. Show that

$$\mathbb{V}[Y] = \mathbb{E}[\mathbb{V}[Y|X]] + \mathbb{V}[\mathbb{E}[Y|X]].$$

Solution: From the first part, setting $Y = W$ and $g_1(X) = g_2(X) = \mathbb{E}[Y]$, it follows that

$$\begin{aligned} \mathbb{V}[Y] &= \mathbb{E}[(Y - \mathbb{E}[Y])^2] \\ &= \mathbb{E}[(Y - \mathbb{E}[Y|X])^2] + \mathbb{E}[(\mathbb{E}[Y|X] - \mathbb{E}[Y])^2] \\ &= \mathbb{E}[\mathbb{E}[(Y - \mathbb{E}[Y|X])^2|X]] + \mathbb{E}[(\mathbb{E}[Y|X] - \mathbb{E}[Y])^2] \\ &= \mathbb{E}[\mathbb{V}[Y|X]] + \mathbb{V}[\mathbb{E}[Y|X]]. \end{aligned}$$

4. Show that

$$\mathbb{C}ov[Y, W] = \mathbb{E}[\mathbb{C}ov[Y, W|X]] + \mathbb{C}ov[\mathbb{E}[Y|X], \mathbb{E}[W|X]],$$

where $\mathbb{C}ov[Y, W|X] = \mathbb{E}[(Y - \mathbb{E}[Y|X])(W - \mathbb{E}[W|X])|X]$.

Solution: From the first part, setting $g_1(X) = \mathbb{E}[Y]$ and $g_2(X) = \mathbb{E}[W]$, the result follows.

5. Show that

$$\mathbb{C}ov[Y, W] \leq \sqrt{\mathbb{E}\big[\mathbb{V}[Y|X]\mathbb{V}[W|X]\big]} + \sqrt{\mathbb{V}\big[\mathbb{E}[Y|X]\big]\mathbb{V}\big[\mathbb{E}[W|X]\big]}.$$

Solution: We have

$$\begin{aligned}\mathbb{C}ov[Y, W] &= \mathbb{E}[\mathbb{C}ov[Y, W|X]] + \mathbb{C}ov[\mathbb{E}[Y|X], \mathbb{E}[W|X]] \\ &\overset{(a)}{\leq} \mathbb{E}\big[\sqrt{\mathbb{V}[Y|X]}\sqrt{\mathbb{V}[W|X]}\big] + \sqrt{\mathbb{V}\big[\mathbb{E}[Y|X]\big]}\sqrt{\mathbb{V}\big[\mathbb{E}[W|X]\big]} \\ &\overset{(b)}{\leq} \sqrt{\mathbb{E}\big[\mathbb{V}[Y|X]\mathbb{V}[W|X]\big]} + \sqrt{\mathbb{V}\big[\mathbb{E}[Y|X]\big]\mathbb{V}\big[\mathbb{E}[W|X]\big]},\end{aligned}$$

where (a) is by the Cauchy-Schwarz inequality for conditional and unconditional expectations, (b) is by $\mathbb{E}|\xi| \leq \sqrt{\mathbb{E}\xi^2}$ for any random variable ξ (which is also the Cauchy-Schwarz inequality applied to ξ and 1).

2 Conservative Inference via Markov Inequality (25 points)

Let $(X_i : 1 \leq i \leq n)$ be pairwise uncorrelated random variables with $\mathbb{E}[X_i] = \mu$ and $\mathbb{V}[X_i] = \sigma^2$ for all $i = 1, \ldots, n$.

1. For each $i = 1, \ldots, n$, and $g \in \mathcal{G}_i = \{g : \mathbb{E}[g(X_i)] < \infty, g \geq 0\}$, show that

$$\mathbb{E}[g(X_i)] \geq \mathbb{E}[g(X_i)\mathbb{1}(g(X_i) > \epsilon)] \geq \epsilon\mathbb{P}[g(X_i) > \epsilon], \qquad \text{for all } \epsilon > 0.$$

Solution: We have

$$\begin{aligned}
\mathbb{E}[g(X_i)] &= \mathbb{E}[g(X_i)\mathbb{1}(g(X_i) \leq \epsilon)] + \mathbb{E}[g(X_i)\mathbb{1}(g(X_i) > \epsilon)] \\
&\geq \mathbb{E}[g(X_i)\mathbb{1}(g(X_i) > \epsilon)] \\
&\geq \epsilon\mathbb{E}[\mathbb{1}(g(X_i) > \epsilon)] = \epsilon\mathbb{P}[g(X_i) > \epsilon]
\end{aligned}$$

2. Show that

$$\mathbb{P}[|\bar{X}_n - \mu| > \epsilon] \leq \frac{\sigma^2}{n\epsilon^2}.$$

Solution: Using the previous part with $g(u) = (u - \mu)^2$, we have

$$\mathbb{P}[|\bar{X}_n - \mu| > \epsilon] \leq \frac{\mathbb{V}[\bar{X}_n]}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2},$$

because $\mathbb{V}[\bar{X}_n] = \mathbb{V}[X_i]/n$.

3. Consider a Wald-type confidence interval for μ of the form

$$I = \left[\bar{X}_n - q \;,\; \bar{X}_n + q\right]$$

for some $q \in \mathbb{R}$. Let $\alpha \in [0, 1]$.

(a) Find a q such that $\mathbb{P}[\mu \in I] \geq 1 - \alpha$.

Solution: Using the result from the previous part,

$$\begin{aligned}
\mathbb{P}[\mu \in I] &= \mathbb{P}[\mu \in [\bar{X}_n - q, \bar{X}_n + q]] \\
&= \mathbb{P}[\bar{X}_n - q \leq \mu \leq \bar{X}_n + q] \\
&= \mathbb{P}[|\bar{X}_n - \mu| \leq q] = 1 - \mathbb{P}[|\bar{X}_n - \mu| > q] \\
&\geq 1 - \frac{\sigma^2}{nq^2}.
\end{aligned}$$

Thus, we set $1 - \alpha = 1 - \frac{\sigma^2}{nq^2}$, and get $q = \sqrt{\frac{\sigma^2}{n\alpha}}$.

(b) Assume that $\{X_i\}$ are independent, distributed $X_i \sim \mathsf{Normal}(\mu, \sigma^2)$, and find a q such that $\mathbb{P}[\mu \in I] = 1 - \alpha$.

Solution: Letting $\Phi(\cdot)$ denote the cumulative distribution of the standard Gaussian distribution,

$$\begin{aligned}
\mathbb{P}[\mu \in I] &= \mathbb{P}[-q \leq \bar{X}_n - \mu \leq q] \\
&= \mathbb{P}\Big[-q\frac{\sqrt{n}}{\sigma} \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq q\frac{\sqrt{n}}{\sigma}\Big] \\
&= \Phi\Big(q\frac{\sqrt{n}}{\sigma}\Big) - \Phi\Big(-q\frac{\sqrt{n}}{\sigma}\Big) \\
&= 1 - 2\Phi\Big(-q\frac{\sqrt{n}}{\sigma}\Big).
\end{aligned}$$

Thus, we set $1 - 2\Phi(-q\frac{\sqrt{n}}{\sigma}) = 1 - \alpha$, to obtain the usual $q = -\frac{\sigma}{\sqrt{n}}\Phi^{-1}(\alpha/2)$.

(c) Compute the interval length of the two confidence intervals constructed above. Which one is better? Discuss, in particular, their difference in light of the role of robustness vs. efficiency in statistical inference.

Solution: The interval lengths for the two proposed confidence intervals are, respectively,

$$2\frac{\sigma}{\sqrt{n}}\frac{1}{\sqrt{\alpha}} \qquad \text{and} \qquad 2\frac{\sigma}{\sqrt{n}}\Phi^{-1}(\alpha/2).$$

Both confidence intervals have the same confidence coefficient because they both satisfy $\inf_{\mu\in\mathbb{R}} \mathbb{P}[\mu \in I] \geq 1 - \alpha$. Thus, in particular, they both control coverage at nominal level $1 - \alpha$. However, their interval lengths differ dramatically, the former being larger than the latter. This reflect the usual robustness-efficiency trade-off, where the first confidence interval is agnostic about the distribution of the data, while the second takes advantage of that information explicitly.

3 M-Estimation with Missing Data (50 points)

Let $(y_i, t_i, \mathbf{x}_i')$, $i = 1, \ldots, n$, be an observed random sample, where $y_i = t_i y_i^*$ with y_i^* taking values in $\mathbb{R}$, t_i taking values $\{0, 1\}$, and $\mathbf{x}_i$ taking values in $\mathbb{R}^d$. The parameter of interest is

$$\theta_0 = \operatorname*{arg\,min}_{\theta \in \Theta} \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)]$$

where $\rho : \mathbb{R}^{d+1} \times \Theta \mapsto \mathbb{R}$ is a smooth loss function, and all necessary moments are assumed to exist.

1. Assume that $(y_i^*, \mathbf{x}_i) \perp\!\!\!\perp t_i$ (i.e., missing completely at random).

   (a) Show that $\mathbb{E}[t_i \rho(y_i, \mathbf{x}_i; \theta)]/p = \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)]$ for all $\theta \in \Theta$, where $p = \mathbb{P}(t_i = 1) > 0$.

   Solution: By independence,

$$\frac{1}{p}\mathbb{E}[t_i \rho(y_i, \mathbf{x}_i; \theta)] = \frac{1}{p}\mathbb{E}[t_i \rho(y_i^*, \mathbf{x}_i; \theta)] = \frac{1}{p}\mathbb{E}[t_i]\mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)] = \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)].$$

   (b) Provide sufficient conditions for $Q_n(\theta) \to_{\mathbb{P}} \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)]$ for all $\theta \in \Theta$, where

$$Q_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \frac{t_i \rho(y_i, \mathbf{x}_i; \theta)}{\widehat{p}_n} \qquad \text{and} \qquad \widehat{p}_n = \frac{1}{n}\sum_{i=1}^{n} t_i.$$

   Solution: By the LLN, $\widehat{p}_n \to_{\mathbb{P}} p$. Thus, by Slutsky's lemma and the continuous mapping theorem,

$$Q_n(\theta) = \frac{p}{\widehat{p}_n}\frac{1}{n}\sum_{i=1}^{n} \frac{t_i \rho(y_i, \mathbf{x}_i; \theta)}{p} \to_{\mathbb{P}} 1 \cdot \mathbb{E}\Big[\frac{t_i \rho(y_i, \mathbf{x}_i; \theta)}{p}\Big] = \mathbb{E}[\rho(y_i^*, \mathbf{x}_i; \theta)].$$

   (c) The previous results motivate the estimator

$$\tilde{\theta}_n = \operatorname*{arg\,min}_{\theta \in \Theta} \frac{1}{n}\sum_{i=1}^{n} \frac{t_i \rho(y_i, \mathbf{x}_i; \theta)}{\widehat{p}_n},$$

   which is consistent for θ_0 under regularity conditions on $\theta \mapsto \rho(\cdot, \cdot; \theta)$.
   Provide sufficient conditions so that

$$\sqrt{n}(\tilde{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \varphi_i + o_{\mathbb{P}}(1) \rightsquigarrow \mathsf{Normal}(0, \Omega_0),$$

   where $(\varphi_i : i = 1, \ldots, n)$ are mean-zero i.i.d., and identify the exact form of φ_i and Ω_0.
   Solution: Assuming $\tilde{\theta}_n \to_{\mathbb{P}} \theta_0$ and $\theta \mapsto \rho(\cdot, \cdot; \theta)$ is twice continuously differentiable,

$$0 = \frac{1}{n}\sum_{i=1}^{n} \frac{t_i}{\widehat{p}_n} m(y_i, \mathbf{x}_i; \tilde{\theta}_n) = \frac{1}{n}\sum_{i=1}^{n} \frac{t_i}{\widehat{p}_n} m(y_i, \mathbf{x}_i; \theta_0) + \Big[\frac{1}{n}\sum_{i=1}^{n} \frac{t_i}{\widehat{p}_n} m_\theta(y_i, \mathbf{x}_i; \check{\theta}_n)\Big](\tilde{\theta}_n - \theta_0),$$

where $m(\cdot,\cdot;\theta) = \frac{\partial}{\partial\theta}\rho(\cdot,\cdot;\theta)$, $m_\theta(\cdot,\cdot;\theta) = \frac{\partial^2}{\partial\theta^2}\rho(\cdot,\cdot;\theta)$, and $\check{\theta}_n = c\tilde{\theta}_n + (1-c)\theta_0$ for some $c \in [0,1]$, and thus $\check{\theta}_n \to_{\mathbb{P}} \theta_0$.

It follows that

$$\sqrt{n}(\tilde{\theta}_n - \theta_0) = -H(\theta_0)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{t_i}{\widehat{p}_n}m(y_i,\mathbf{x}_i;\theta_0) + o_{\mathbb{P}}(1),$$

provided that

$$H_n(\check{\theta}_n) = \frac{1}{n}\sum_{i=1}^{n}\frac{t_i}{\widehat{p}_n}m_\theta(y_i,\mathbf{x}_i;\check{\theta}_n) \to_{\mathbb{P}} \mathbb{E}\Big[\frac{t_i}{p}m_\theta(y_i,\mathbf{x}_i;\theta_0)\Big] = \mathbb{E}[m_\theta(y_i^*,\mathbf{x}_i;\theta_0)] = H(\theta_0)$$

and

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{t_i}{\widehat{p}_n}m(y_i,\mathbf{x}_i;\theta_0) = O_{\mathbb{P}}(1).$$

For the first requirement we have

$$|H_n(\check{\theta}_n) - H(\theta_0)| \leq |H_n(\check{\theta}_n) - H(\check{\theta}_n)| + |H(\check{\theta}_n) - H(\theta_0)| = o_{\mathbb{P}}(1) + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1),$$

provided that $\sup_{\theta\in\Theta_\delta}|H_n(\theta) - H(\theta)| = o_{\mathbb{P}}(1)$, where $\Theta_\delta = \{\theta \in \Theta : |\theta - \theta_0| \leq \delta\}$ for some $\delta > 0$ (uniform law of large numbers), and $|H(\check{\theta}_n) - H(\theta_0)| = o_{\mathbb{P}}(1)$ (continuous mapping theorem). The second requirement, if $\mathbb{E}[|m(y_i^*,\mathbf{x}_i;\theta_0)|^2] < \infty$, by the CLT,

$$\begin{aligned}
&\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{t_i}{\widehat{p}_n}m(y_i,\mathbf{x}_i;\theta_0) \\
&\quad= \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{t_i m(y_i,\mathbf{x}_i;\theta_0)}{p} - \sqrt{n}\frac{\widehat{p}-p}{\widehat{p}p}\frac{1}{n}\sum_{i=1}^{n}t_i m(y_i,\mathbf{x}_i;\theta_0) \\
&\quad= \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{t_i m(y_i^*,\mathbf{x}_i;\theta_0)}{p} + \sqrt{n}O_{\mathbb{P}}(n^{-1/2})O_{\mathbb{P}}(n^{-1/2}) = O_{\mathbb{P}}(1),
\end{aligned}$$

since $\mathbb{E}[t_i m(y_i^*,\mathbf{x}_i;\theta_0)] = p\mathbb{E}[m(y_i^*,\mathbf{x}_i;\theta_0)] = 0$.

Therefore, we have

$$\varphi_i = -H(\theta_0)^{-1}\frac{t_i m(y_i^*,\mathbf{x}_i;\theta_0)}{p}$$

and

$$\begin{aligned}
\Omega_0 = \mathbb{V}[\varphi_i] &= H(\theta_0)^{-1}\mathbb{V}\Big[\frac{t_i}{p}m(y_i^*,\mathbf{x}_i;\theta_0)\Big]H(\theta_0)^{-1} \\
&= H(\theta_0)^{-1}\frac{1}{p}\mathbb{E}[m(y_i^*,\mathbf{x}_i;\theta_0)^2]H(\theta_0)^{-1}.
\end{aligned}$$

(d) Propose a consistent estimator of Ω_0, and show it is consistent.

Solution: A consistent estimator of Ω_0 is

$$\widehat{\Omega} = H_n(\tilde{\theta}_n)^{-1}\frac{1}{\widehat{p}_n}\Big[\frac{1}{n}\sum_{i=1}^{n}\frac{t_i}{\widehat{p}_n}(m(y_i,\mathbf{x}_i;\tilde{\theta}_n))^2\Big]H_n(\tilde{\theta}_n)^{-1}.$$

Proceeding exactly as above, it follows that $H_n(\tilde{\theta}_n) \to_{\mathbb{P}} H(\theta_0)$. Similarly,

$$\begin{aligned}
&\frac{1}{n}\sum_{i=1}^{n}\frac{t_i}{\widehat{p}_n}(m(y_i,\mathbf{x}_i;\tilde{\theta}_n))^2\\
&=\frac{1}{n}\sum_{i=1}^{n}\frac{t_i}{\widehat{p}_n}(m(y_i^*,\mathbf{x}_i;\theta_0))^2+\frac{1}{n}\sum_{i=1}^{n}\frac{t_i}{\widehat{p}_n}\big[(m(y_i^*,\mathbf{x}_i;\tilde{\theta}_n))^2-(m(y_i^*,\mathbf{x}_i;\theta_0))^2\big]\\
&=\mathbb{E}\Big[\frac{t_i}{p}(m(y_i^*,\mathbf{x}_i;\theta_0))^2\Big]+o_{\mathbb{P}}(1),
\end{aligned}$$

because, for example, we can assume that $|(m(y_i^*,\mathbf{x}_i;\tilde{\theta}_n))^2-(m(y_i^*,\mathbf{x}_i;\theta_0))^2| \le C|\tilde{\theta}_n - \theta_0| = o_{\mathbb{P}}(1)$. Finally, because $\widehat{p}_n \to p$, it follows by Slutsky's lemma and the continuous mapping theorem that $\widehat{\Omega} \to_{\mathbb{P}} \Omega_0$.

(e) Construct an asymptotically valid feasible 95% confidence interval for θ.

Solution: An asymptotic valid 95% confidence interval for θ is

$$\Big[\ \tilde{\theta} - \Phi^{-1}_{1-\alpha/2}\sqrt{\widehat{\Omega}/n}\ ,\ \tilde{\theta} + \Phi^{-1}_{1-\alpha/2}\sqrt{\widehat{\Omega}/n}\ \Big]$$

where $\Phi^{-1}_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of standard normal and α is set to be 0.05.

2. Assume that $y_i^* \perp\!\!\!\perp t_i \mid \mathbf{x}_i$ (i.e., conditional independence, unconfoundedness).

(a) Show that $\mathbb{E}[t_i\rho(y_i,\mathbf{x}_i;\theta)/p(\mathbf{x}_i)] = \mathbb{E}[\rho(y_i^*,\mathbf{x}_i;\theta)]$ for all $\theta \in \Theta$, where $p(\mathbf{x}_i) = \mathbb{P}(t_i = 1|\mathbf{x}_i) > 0$.

Solution: By the conditional independence assumption,

$$\begin{aligned}
\mathbb{E}\Big[\frac{t_i\rho(y_i,\mathbf{x}_i;\theta)}{p(\mathbf{x}_i)}\Big] &= \mathbb{E}\Big[\frac{\mathbb{E}[t_i\rho(y_i^*,\mathbf{x}_i;\theta)|\mathbf{x}_i]}{p(\mathbf{x}_i)}\Big]\\
&= \mathbb{E}\Big[\frac{\mathbb{E}[t_i|\mathbf{x}_i]\mathbb{E}[\rho(y_i^*,\mathbf{x}_i;\theta)|\mathbf{x}_i]}{p(\mathbf{x}_i)}\Big] = \mathbb{E}[\mathbb{E}[\rho(y_i^*,\mathbf{x}_i;\theta)|\mathbf{x}_i]].
\end{aligned}$$

(b) Assuming $\max_{1\le i\le n}|\widehat{p}_n(\mathbf{x}_i) - p(\mathbf{x}_i)| = o_{\mathbb{P}}(1)$ and $\min_{1\le i\le n} p(\mathbf{x}_i) \ge p_{\min} > 0$, provide sufficient conditions for $M_n(\theta) \to_{\mathbb{P}} \mathbb{E}[\rho(y_i^*,\mathbf{x}_i;\theta)]$ for all $\theta \in \Theta$, where

$$M_n(\theta) = \frac{1}{n}\sum_{i=1}^{n}\frac{t_i\rho(y_i,\mathbf{x}_i;\theta)}{\widehat{p}_n(\mathbf{x}_i)}.$$

Solution: We have

$$\begin{aligned} M_n(\theta) &= \frac{1}{n}\sum_{i=1}^n \frac{t_i\rho(y_i,\mathbf{x}_i;\theta)}{p(\mathbf{x}_i)} + \frac{1}{n}\sum_{i=1}^n \Big(\frac{1}{\widehat{p}_n(\mathbf{x}_i)} - \frac{1}{p(\mathbf{x}_i)}\Big) t_i\rho(y_i,\mathbf{x}_i;\theta) \\ &= \frac{1}{n}\sum_{i=1}^n \frac{t_i\rho(y_i,\mathbf{x}_i;\theta)}{p(\mathbf{x}_i)} + o_{\mathbb{P}}(1) = \mathbb{E}\Big[\frac{t_i\rho(y_i,\mathbf{x}_i;\theta)}{p(\mathbf{x}_i)}\Big] + o_{\mathbb{P}}(1) = \mathbb{E}[\rho(y_i^*,\mathbf{x}_i;\theta)] + o_{\mathbb{P}}(1), \end{aligned}$$

by the LLN, and because (for n large enough)

$$\Big|\frac{1}{n}\sum_{i=1}^n \Big(\frac{t_i\rho(y_i,\mathbf{x}_i;\theta)}{\widehat{p}_n(\mathbf{x}_i)} - \frac{t_i\rho(y_i,\mathbf{x}_i;\theta)}{p(\mathbf{x}_i)}\Big)\Big| \leq \max_{1\leq i\leq n} |\widehat{p}_n(\mathbf{x}_i) - p(\mathbf{x}_i)| \frac{1}{n}\sum_{i=1}^n \frac{t_i|\rho(y_i^*,\mathbf{x}_i;\theta)|}{p(\mathbf{x}_i)|\widehat{p}_n(\mathbf{x}_i)|} = o_{\mathbb{P}}(1),$$

using the condition $\min_{1\leq i\leq n} p(\mathbf{x}_i) \geq p_{\min} > 0$.

(c) The previous results motivate the estimator

$$\widehat{\theta}_n = \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^n \frac{\rho(y_i,\mathbf{x}_i;\theta)}{\widehat{p}_n(\mathbf{x}_i)},$$

which is consistent for θ_0 under regularity conditions on $\theta \mapsto \rho(\cdot,\cdot;\theta)$ and the (first-step) estimator $\widehat{p}_n(\cdot)$. In particular, to simplify the problem, assume that $\widehat{p}_n(\mathbf{x}) = p(\mathbf{x};\widehat{\boldsymbol{\gamma}})$, where $\gamma \mapsto p(\cdot;\boldsymbol{\gamma})$ is a known smooth function, and the parametric estimator $\widehat{\boldsymbol{\gamma}}$ admits an asymptotic linear representation as follows:

$$\sqrt{n}(\widehat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^n \vartheta_i + o_{\mathbb{P}}(1), \quad \text{where } \{\vartheta_i\}_{i=1}^n \text{ i.i.d. with } \mathbb{E}[\vartheta_i]=0,\ \mathbb{V}[\vartheta_i]<\infty.$$

Provide sufficient conditions so that

$$\sqrt{n}(\widehat{\theta}_n - \theta) = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_i + o_{\mathbb{P}}(1) \rightsquigarrow \mathsf{Normal}(0,\Sigma_0),$$

where $\{\psi_i\}_{i=1}^n$ are mean-zero i.i.d., and identify the exact form of ψ_i and Σ_0.

Solution: Assuming $\widehat{\theta}_n \to_{\mathbb{P}} \theta_0$ and $\theta \mapsto \rho(\cdot,\cdot;\theta)$ is twice continuously differentiable,

$$\begin{aligned} 0 &= \frac{1}{n}\sum_{i=1}^n \frac{t_i}{\widehat{p}_n(\mathbf{x}_i)} m(y_i,\mathbf{x}_i;\widehat{\theta}_n) \\ &= \frac{1}{n}\sum_{i=1}^n \frac{t_i}{\widehat{p}_n(\mathbf{x}_i)} m(y_i,\mathbf{x}_i;\theta_0) + \Big[\frac{1}{n}\sum_{i=1}^n \frac{t_i}{\widehat{p}_n(\mathbf{x}_i)} m_\theta(y_i,\mathbf{x}_i;\check{\theta}_n)\Big](\widehat{\theta}_n - \theta_0), \end{aligned}$$

where $m(\cdot,\cdot;\theta) = \frac{\partial}{\partial\theta}\rho(\cdot,\cdot;\theta)$, $m_\theta(\cdot,\cdot;\theta) = \frac{\partial^2}{\partial\theta^2}\rho(\cdot,\cdot;\theta)$, and $\check{\theta}_n = c\tilde{\theta}_n + (1-c)\theta_0$ for some $c\in[0,1]$. Thus, $\check{\theta}_n \to_{\mathbb{P}} \theta_0$.

Next, we have

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{t_i}{\widehat{p}_n(\mathbf{x}_i)}m(y_i,\mathbf{x}_i;\theta_0)=T_1+T_2+T_3,$$

where

$$\begin{aligned}
T_1&=\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{t_i}{p(\mathbf{x}_i)}m(y_i,\mathbf{x}_i;\theta_0)\\
T_2&=-\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{\widehat{p}_n(\mathbf{x}_i)-p(\mathbf{x}_i)}{p(\mathbf{x}_i)^2}t_im(y_i,\mathbf{x}_i;\theta_0)\\
T_3&=\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{(\widehat{p}_n(\mathbf{x}_i)-p(\mathbf{x}_i))^2}{\widehat{p}_n(\mathbf{x}_i)p(\mathbf{x}_i)^2}t_im(y_i,\mathbf{x}_i;\theta_0).
\end{aligned}$$

We have

$$\sqrt{n}(\widehat{p}_n(\mathbf{x}_i)-p(\mathbf{x}_i))=\sqrt{n}(p(\mathbf{x}_i;\widehat{\boldsymbol{\gamma}})-p(\mathbf{x}_i;\boldsymbol{\gamma}_0))=p_\gamma(\mathbf{x}_i;\check{\boldsymbol{\gamma}})\sqrt{n}(\widehat{\boldsymbol{\gamma}}-\boldsymbol{\gamma}_0),$$

where $p_\gamma(\cdot;\boldsymbol{\gamma})=\frac{\partial}{\partial\boldsymbol{\gamma}}\rho(\cdot;\boldsymbol{\gamma})$, and $\check{\boldsymbol{\gamma}}_n=c\widehat{\boldsymbol{\gamma}}_n+(1-c)\boldsymbol{\gamma}_0$ for some $c\in[0,1]$, and thus $\check{\boldsymbol{\gamma}}_n\to_{\mathbb{P}}\boldsymbol{\gamma}_0$. Since $\widehat{\boldsymbol{\gamma}}$ admits an asymptotic linear representation, it follows that $\max_{1\le i\le n}|\widehat{p}_n(\mathbf{x}_i)-p(\mathbf{x}_i)|=O_{\mathbb{P}}(n^{-1/2})$ and $|p_\gamma(\mathbf{x}_i;\check{\boldsymbol{\gamma}})-p_\gamma(\mathbf{x}_i;\boldsymbol{\gamma}_0)|=o_{\mathbb{P}}(1)$, provided that $\gamma\mapsto p(\cdot;\boldsymbol{\gamma})$ and $\gamma\mapsto p_\gamma(\cdot;\boldsymbol{\gamma})$ are continuously differentiable and uniformly bounded.

Therefore,

$$\begin{aligned}
T_2&=-\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{\widehat{p}_n(\mathbf{x}_i)-p(\mathbf{x}_i)}{p(\mathbf{x}_i)^2}t_im(y_i,\mathbf{x}_i;\theta_0)\\
&=-\Big[\frac{1}{n}\sum_{i=1}^{n}\frac{p_\gamma(\mathbf{x}_i;\check{\boldsymbol{\gamma}})}{p(\mathbf{x}_i)^2}t_im(y_i,\mathbf{x}_i;\theta_0)\Big]\sqrt{n}(\widehat{\boldsymbol{\gamma}}-\boldsymbol{\gamma}_0)\\
&=-\Big[\frac{1}{n}\sum_{i=1}^{n}\frac{p_\gamma(\mathbf{x}_i;\boldsymbol{\gamma}_0)}{p(\mathbf{x}_i)^2}t_im(y_i,\mathbf{x}_i;\theta_0)\Big]\sqrt{n}(\widehat{\boldsymbol{\gamma}}-\boldsymbol{\gamma}_0)+o_{\mathbb{P}}(1)\\
&=-\mathbb{E}\Big[\frac{p_\gamma(\mathbf{x}_i;\boldsymbol{\gamma}_0)}{p(\mathbf{x}_i)}m(y_i^*,\mathbf{x}_i;\theta_0)\Big]\sqrt{n}(\widehat{\boldsymbol{\gamma}}-\boldsymbol{\gamma}_0)+o_{\mathbb{P}}(1)\\
&=-\frac{1}{\sqrt{n}}\sum_{i=1}^{n}G(\theta_0,\boldsymbol{\gamma}_0)\vartheta_i+o_{\mathbb{P}}(1)
\end{aligned}$$

where

$$G(\theta_0,\boldsymbol{\gamma}_0)=\mathbb{E}\Big[\frac{p_\gamma(\mathbf{x}_i;\boldsymbol{\gamma}_0)}{p(\mathbf{x}_i)}m(y_i^*,\mathbf{x}_i;\theta_0)\Big].$$

Furthermore,

$$
\begin{aligned}
|T_3| &\leq \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{(\widehat{p}_n(\mathbf{x}_i) - p(\mathbf{x}_i))^2}{|\widehat{p}_n(\mathbf{x}_i)| p(\mathbf{x}_i)^2} |m(y_i^*, \mathbf{x}_i; \theta_0)| \\
&= \sqrt{n} \max_{1 \leq i \leq n} |\widehat{p}_n(\mathbf{x}_i) - p(\mathbf{x}_i)| O_{\mathbb{P}}(1) = o_{\mathbb{P}}(1).
\end{aligned}
$$

Finally, proceeding as above, we can show that

$$
H_n(\check{\theta}_n) = \frac{1}{n} \sum_{i=1}^{n} \frac{t_i}{\widehat{p}_n(\mathbf{x}_i)} m_\theta(y_i, \mathbf{x}_i; \check{\theta}_n) \to_{\mathbb{P}} \mathbb{E}\Big[\frac{t_i}{p(\mathbf{x}_i)} m_\theta(y_i, \mathbf{x}_i; \theta_0)\Big] = \mathbb{E}[m_\theta(y_i^*, \mathbf{x}_i; \theta_0)] = H(\theta_0).
$$

Putting the results together,

$$
\psi_i = -H(\theta_0)^{-1} \Big[\frac{t_i m(y_i^*, \mathbf{x}_i; \theta_0)}{p(\mathbf{x}_i)} - G(\theta_0, \boldsymbol{\gamma}_0) \vartheta_i\Big]
$$

and

$$
\Sigma_0 = H(\theta_0)^{-1} \mathbb{V}\Big[\frac{t_i m(y_i^*, \mathbf{x}_i; \theta_0)}{p(\mathbf{x}_i)} - G(\theta_0, \boldsymbol{\gamma}_0) \vartheta_i\Big] H(\theta_0)^{-1}.
$$

(d) Propose a consistent estimator of Σ_0.

Solution: Proceeding as above, we have $|H_n(\widehat{\theta}_n) - H(\theta_0)| \to_{\mathbb{P}} 0$. Furthermore, it also follows that

$$
\begin{aligned}
\widehat{G}(\widehat{\theta}_n, \widehat{\boldsymbol{\gamma}}_n) &= \frac{1}{n} \sum_{i=1}^{n} \frac{p_\gamma(\mathbf{x}_i; \widehat{\boldsymbol{\gamma}}_n) t_i m(y_i, \mathbf{x}_i; \widehat{\theta}_n)}{p(\mathbf{x}_i; \widehat{\boldsymbol{\gamma}}_n)^2} \\
&= \frac{1}{n} \sum_{i=1}^{n} \frac{p_\gamma(\mathbf{x}_i; \boldsymbol{\gamma}_0) t_i m(y_i, \mathbf{x}_i; \theta_0)}{p(\mathbf{x}_i; \boldsymbol{\gamma}_0)^2} + o_{\mathbb{P}}(1) \\
&= \mathbb{E}\Big[\frac{p_\gamma(\mathbf{x}_i; \boldsymbol{\gamma}_0) t_i m(y_i, \mathbf{x}_i; \theta_0)}{p(\mathbf{x}_i; \boldsymbol{\gamma}_0)^2}\Big] + o_{\mathbb{P}}(1) \\
&= \mathbb{E}\Big[\frac{p_\gamma(\mathbf{x}_i; \boldsymbol{\gamma}_0)}{p(\mathbf{x}_i)} m(y_i^*, \mathbf{x}_i; \theta_0)\Big] + o_{\mathbb{P}}(1) \\
&= G(\theta_0, \boldsymbol{\gamma}_0) + o_{\mathbb{P}}(1).
\end{aligned}
$$

Furthermore,

$$
\mathbb{V}\Big[\frac{t_i m(y_i^*, \mathbf{x}_i; \theta_0)}{p(\mathbf{x}_i)} - G(\theta_0, \boldsymbol{\gamma}_0) \vartheta_i\Big] = \mathbb{E}\Big[\Big(\frac{t_i m(y_i, \mathbf{x}_i; \theta_0)}{p(\mathbf{x}_i)} - G(\theta_0, \boldsymbol{\gamma}_0) \vartheta_i\Big)^2\Big],
$$

and hence we have the following plug-in estimator

$$
\widehat{\mathbb{E}}\Big[\Big(\frac{t_i m(y_i, \mathbf{x}_i; \widehat{\theta}_n)}{p(\mathbf{x}_i; \widehat{\boldsymbol{\gamma}}_n)} - \widehat{G}(\widehat{\theta}_n, \widehat{\boldsymbol{\gamma}}_n) \widehat{\vartheta}_i\Big)^2\Big]
$$

$$= \frac{1}{n} \sum_{i=1}^{n} \Big(\frac{t_i m(y_i, \mathbf{x}_i; \widehat{\theta}_n)}{p(\mathbf{x}_i; \widehat{\boldsymbol{\gamma}}_n)} - \widehat{G}(\widehat{\theta}_n, \widehat{\boldsymbol{\gamma}}_n) \widehat{\vartheta}_i \Big)^2,$$

which can be shown to be consistent as before, provided (among other things) that $\max_{1 \le i \le n} |\widehat{\vartheta}_i - \vartheta_i| = o_{\mathbb{P}}(1)$.

(e) Construct an asymptotically valid 95% confidence interval for θ_0.

Solution: An asymptotic valid 95% confidence interval for θ is

$$\Big[\, \tilde{\theta} - \Phi^{-1}_{1-\alpha/2} \sqrt{\widehat{\Sigma}/n} \,,\ \tilde{\theta} + \Phi^{-1}_{1-\alpha/2} \sqrt{\widehat{\Sigma}/n} \,\Big]$$

where $\Phi^{-1}_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of standard normal and α is set to be 0.05.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m,\ x = a_1, ..., a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m,\ \ t \in \mathcal{R}$
$DU(a_1, ..., a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m,\ \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R},\ m = 1, 2, ...$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x},\ \ x = 0, 1, ..., n$
	m.g.f.	$(pe^t + 1 - p)^n,\ \ t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1],\ \ n = 1, 2, ...$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!,\ \ x = 0, 1, 2, ...$
	m.g.f.	$e^{\theta(e^t - 1)},\ \ t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p,\ \ x = 1, 2, ...$
	m.g.f.	$pe^t/[1 - (1-p)e^t],\ \ t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-	p.d.f.	$\binom{n}{x}\binom{m}{r-x} / \binom{N}{r}$
geometric		$x = 0, 1, ..., \min\{r, n\},\ \ r - x \le m$
	m.g.f.	No explicit form
$HG(r, n, m)$	Expectation	rn/N
	Variance	$rnm(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, ...,\ N = n + m$
Negative	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r},\ \ x = r, r+1, ...$
binomial	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r,\ \ t < -\log(1-p)$
	Expectation	r/p
$NB(p, r)$	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1],\ \ r = 1, 2, ...$
Log-	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x,\ \ x = 1, 2, ...$
distribution	m.g.f.	$\log[1 - (1-p)e^t]/\log p,\ \ t \in \mathcal{R}$
	Expectation	$-(1-p)/(p \log p)$
$L(p)$	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt}-e^{at})/[(b-a)t],\ \ t\in\mathcal{R}$
$U(a,b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a,\ b\in\mathcal{R},\ \ a<b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t+\sigma^2t^2/2},\ \ t\in\mathcal{R}$
$N(\mu,\sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1},\ \ t<\theta^{-1}$
$E(a,\theta)$	Expectation	$\theta+a$
	Variance	θ^2
	Parameter	$\theta>0,\ \ a\in\mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2},\ \ t<1/2$
χ^2_k	Expectation	k
	Variance	$2k$
	Parameter	$k=1,2,...$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha},\ \ t<\gamma^{-1}$
$\Gamma(\alpha,\gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma>0,\ \alpha>0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha,\beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha>0,\ \ \beta>0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t-\sigma\|t\|}$
$C(\mu,\sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)}\left(1+\frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	$0, \ (n > 1)$
	Variance	$n/(n-2), \ (n > 2)$
	Parameter	$n = 1, 2, ...$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2), \ (m > 2)$
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, $(m > 4)$
	Parameter	$n = 1, 2, ..., \ m = 1, 2, ...$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x-\mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2}-1)$
	Parameter	$\mu \in \mathcal{R}, \ \sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1}+1)$
	Variance	$\theta^{2/\alpha}\left\{\Gamma(2\alpha^{-1}+1)-\left[\Gamma(\alpha^{-1}+1)\right]^2\right\}$
	Parameter	$\theta > 0, \ \alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{-\lvert x-\mu\rvert/\theta}$
	m.g.f.	$e^{\mu t}/(1-\theta^2t^2), \ \lvert t\rvert < \theta^{-1}$
	Expectation	μ
$DE(\mu, \theta)$	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}, \ \theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta-1), \ (\theta > 1)$
	Variance	$\theta a^2/[(\theta-1)^2(\theta-2)], \ (\theta > 2)$
	Parameter	$\theta > 0, \ a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1+e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1+\sigma t)\Gamma(1-\sigma t), \ \lvert t\rvert < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}, \ \sigma > 0$